(9.9) $\vec{E}(\vec{r},t) = E_0 \cos(\vec{k}\cdot\vec{r} - \omega t + \delta)\,\hat{n}$

$\vec{B}(\vec{r},t) = \frac{1}{c} E_0 \cos(\vec{k}\cdot\vec{r} - \omega t + \delta)\,(\hat{k}\times\hat{n})$

always $k = |\vec{k}| = \frac{\omega}{c}$

$\delta = 0$

(a) $\vec{k} = -\frac{\omega}{c}\hat{x} \Rightarrow \vec{k}\cdot\vec{r} = -\frac{\omega}{c}x$

$\hat{n} = \hat{z} \Rightarrow \hat{k}\times\hat{n} = -\hat{x}\times\hat{z} = \hat{y}$

$$\Rightarrow \vec{E} = E_0 \cos\left(\frac{\omega}{c}x + \omega t\right)\hat{z}$$

$$\vec{B} = \frac{1}{c} E_0 \cos\left(\frac{\omega}{c}x + \omega t\right)\hat{y}$$

(b) $\vec{k} = \frac{\omega}{c}\,\frac{\hat{x}+\hat{y}+\hat{z}}{\sqrt{3}}$ so that $\hat{k} = \frac{1}{\sqrt{3}}(1,1,1)$ is unit vector.

Now $\hat{n} \in xz$ plane $\Rightarrow \hat{n} = \alpha\hat{x} + \beta\hat{z}$

But need to make sure that $\hat{k}\cdot\hat{n} = 0 \Rightarrow \alpha = -\beta$

$\Rightarrow$ normalizing $\hat{n} = \frac{1}{\sqrt{2}}(\hat{x} - \hat{z})$

Then $\hat{k}\times\hat{n} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{vmatrix} = \hat{x}\left(-\frac{1}{\sqrt{6}}\right) - \hat{y}\left(-\frac{1}{\sqrt{6}} - \frac{1}{\sqrt{6}}\right) + \hat{z}\left(-\frac{1}{\sqrt{6}}\right) =$

$$= \frac{-1}{\sqrt{6}}\left(\hat{x} - 2\hat{y} + \hat{z}\right)$$

$$\vec{E} = E_0 \cos\left(\frac{\omega}{\sqrt{3}c}(x+y+z) - \omega t\right)\frac{\hat{x}-\hat{z}}{\sqrt{2}}$$

$$\vec{B} = \frac{1}{c} E_0 \cos\left(\frac{\omega}{\sqrt{3}c}(x+y+z) - \omega t\right)\frac{-\hat{x}+2\hat{y}-\hat{z}}{\sqrt{6}}$$

(9.13) $$R = \left(\frac{E_{0R}}{E_{0i}}\right)^2 = \left(\frac{1-\beta}{1+\beta}\right)^2 \quad \text{from (9.82)}$$

$$T = \frac{\varepsilon_2 v_2}{\varepsilon_1 v_1}\left(\frac{E_{0T}}{E_{0i}}\right)^2 = \frac{\varepsilon_2 v_2}{\varepsilon_1 v_1}\left(\frac{2}{1+\beta}\right)^2 \quad \text{from (9.82)}$$

$$\beta = \frac{\mu_1 v_1}{\mu_2 v_2}.$$

Note that
$$\left.\begin{aligned} \varepsilon_2 v_2 &= \frac{c}{n_2} = c\sqrt{\frac{\varepsilon_0\mu_0}{\varepsilon_2\mu_2}} \\ \varepsilon_1 v_1 &= \frac{c}{n_1} = c\sqrt{\frac{\varepsilon_0\mu_0}{\varepsilon_1\mu_1}} \end{aligned}\right\} \quad \frac{\varepsilon_2 v_2}{\varepsilon_1 v_1} = \sqrt{\frac{\varepsilon_1\mu_1}{\varepsilon_2\mu_2}}$$

$$\frac{\varepsilon_2 v_2}{\varepsilon_1 v_1} = \frac{\varepsilon_2\mu_2 v_2}{\varepsilon_1\mu_1 v_1}\frac{\mu_1}{\mu_2} = \left(\frac{v_1}{v_2}\right)^2\frac{v_2}{v_1}\frac{\mu_1}{\mu_2} = \frac{v_1\mu_1}{v_2\mu_2} = \beta$$

but $\dfrac{\varepsilon_2\mu_2}{\varepsilon_1\mu_1} = \left(\dfrac{v_1}{v_2}\right)^2$ from (9.68)

$$\Rightarrow T = \beta\left(\frac{2}{1+\beta}\right)^2$$

$$\Rightarrow T + R = \frac{(1-\beta)^2 + 4\beta}{(1+\beta)^2} = \frac{1+2\beta+\beta^2}{(1+\beta)^2} = 1.$$

(9.19) This problem was mostly done in class. But briefly: from (9.126)

$$\varkappa = \omega\sqrt{\frac{\varepsilon\mu}{2}}\left[\sqrt{1+\left(\frac{\sigma}{\varepsilon\omega}\right)^2} - 1\right]^{1/2}$$

(a) $\sigma \ll \omega\varepsilon \Rightarrow$ expand square root $\sqrt{1+\left(\frac{\sigma}{\varepsilon\omega}\right)^2} - 1 \approx \frac{1}{2}\left(\frac{\sigma}{\varepsilon\omega}\right)^2$

$$\varkappa = \omega\sqrt{\frac{\varepsilon\mu}{2}}\sqrt{\frac{1}{2}\left(\frac{\sigma}{\varepsilon\omega}\right)^2} = \frac{1}{2}\sigma\sqrt{\frac{\mu}{\varepsilon}} \Rightarrow d = \frac{1}{\varkappa} = \frac{2}{\sigma}\sqrt{\frac{\varepsilon}{\mu}}$$

For water $\varepsilon = \varepsilon_r \varepsilon_0 \cong 80\, \varepsilon_0$ (p. 180) $\varepsilon_0 = 8.85 \cdot 10^{-12}$

$\mu = \mu_0 (1 + \chi_m) \approx \mu_0$ (p. 275) $\mu_0 = 4\pi \cdot 10^{-7}$

$\sigma = \frac{1}{\rho} = \frac{1}{2.5 \cdot 10^5}$ (p. 286)

$$d = 2\,(2.5 \cdot 10^5) \sqrt{\frac{80\, \varepsilon_0}{\mu_0}} \approx 1.2 \cdot 10^4 \text{ m.}$$

(b) good conductor $\sigma \gg \varepsilon\omega$

$$\Rightarrow \varkappa \cong \omega \sqrt{\frac{\varepsilon\mu}{2}} \left(\frac{\sigma}{\varepsilon\omega}\right)^{1/2} = \sqrt{\frac{\omega\mu\sigma}{2}}$$

$$d = \sqrt{\frac{2}{\omega\mu\sigma}} \approx 1.3 \cdot 10^{-8} \text{ m} \quad \text{for } \sigma = 10^7 \frac{1}{\Omega \cdot \text{m}}$$

$\omega \approx 10^{15}\, s^{-1}$

$\mu = \mu_0 = 4\pi \cdot 10^{-7}$

(c) Real part of $\tilde{k}$ is $k = \omega \sqrt{\frac{\varepsilon\mu}{2}} \left(\sqrt{1 + \left(\frac{\sigma}{\varepsilon\omega}\right)^2} + 1 \right)^{1/2} \approx$

$$\approx \sqrt{\frac{\omega\mu\sigma}{2}} \quad \text{for } \sigma \gg \varepsilon\omega$$

$$\Rightarrow \text{eq. 9.134: } \phi = \tan^{-1}\left(\frac{\varkappa}{k}\right) = \tan^{-1}(1) = \frac{\pi}{4}.$$

$$\frac{B_0}{E_0} = \left[\varepsilon\mu \sqrt{1 + \left(\frac{\sigma}{\varepsilon\omega}\right)^2} \right]^{1/2} \approx \left(\varepsilon\mu \frac{\sigma}{\varepsilon\omega} \right)^{1/2} = \sqrt{\frac{\sigma\mu}{\omega}} \approx 10^{-7}$$

while in vacuum $\frac{B_0}{E_0} = \frac{1}{c} \approx 3 \cdot 10^{-9}\, s/m$

$\Rightarrow$ mag. field is relatively stronger in metals.

(9.25) $\gamma_j = 0 \quad \vec{E} = E_0 e^{-\varkappa z} e^{i(kz-\omega t)}$

$$k = \frac{\omega}{c}\left[1 + \frac{Nq^2}{2m\varepsilon_0}\sum_j \frac{f_j}{\omega_j^2-\omega^2}\right]$$

$$v_g = \frac{d\omega}{dk} = \frac{1}{dk/d\omega}$$

$$\Rightarrow \frac{dk}{d\omega} = \frac{1}{c}\left[1 + \frac{Nq^2}{2m\varepsilon_0}\sum \frac{f_j}{\omega_j^2-\omega^2}\right] + \frac{\omega}{c}\,\frac{Nq^2}{2m\varepsilon_0}\sum_j \frac{f_j \cdot 2\omega}{(\omega_j^2-\omega^2)^2}$$

$$= \frac{1}{c}\left[1 + \frac{Nq^2}{2m\varepsilon_0}\sum_j \frac{f_j}{\omega_j^2-\omega^2} + \frac{Nq^2}{2m\varepsilon_0}\sum_j \frac{f_j \cdot 2\omega^2}{(\omega_j^2-\omega^2)^2}\right] =$$

$$= \frac{1}{c}\left[1 + \frac{Nq^2}{2m\varepsilon_0}\sum_j f_j \frac{\omega_j^2-\omega^2+2\omega^2}{(\omega_j^2-\omega^2)^2}\right] =$$

$$= \frac{1}{c}\left\{1 + \frac{Nq^2}{2m\varepsilon_0}\sum_j f_j \frac{\omega_j^2+\omega^2}{(\omega_j^2-\omega^2)^2}\right\}.$$

$$\Rightarrow v_g = c\left\{1 + \frac{Nq^2}{2m\varepsilon_0}\sum_j f_j \frac{\omega_j^2+\omega^2}{(\omega_j^2-\omega^2)^2}\right\}^{-1}$$

But the bracuets are greater than 1,

hence $v_g < c \Rightarrow$ group velocity is smaller than speed of light.

Note that Phase velocity $v = \frac{\omega}{k} = c\left\{1 + \frac{Nq^2}{2m\varepsilon_0}\sum_j \frac{f_j}{\omega_j^2-\omega^2}\right\} > c$

for $\omega < \omega_j$

P9.17

Problem 9.17

Equation 9.106 $\Rightarrow \beta = 2.42$; Eq. 9.110 $\Rightarrow$

$$\alpha = \frac{\sqrt{1-(\sin\theta/2.42)^2}}{\cos\theta}.$$

(a) $\theta = 0 \Rightarrow \alpha = 1$. Eq. 9.109 $\Rightarrow \left(\frac{E_{0_R}}{E_{0_I}}\right) = \frac{\alpha-\beta}{\alpha+\beta} =$

$$\frac{1-2.42}{1+2.42} = -\frac{1.42}{3.42} = \boxed{-0.415;}$$

$$\left(\frac{E_{0_T}}{E_{0_I}}\right) = \frac{2}{\alpha+\beta} = \frac{2}{3.42} = \boxed{0.585.}$$

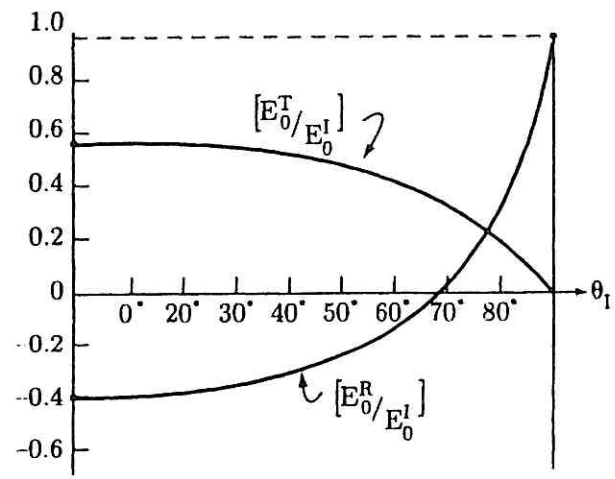
(b) Equation 9.112 $\Rightarrow \theta_B = \tan^{-1}(2.42) = \boxed{67.5°.}$

(c) $E_{0_R} = E_{0_T} \Rightarrow \alpha - \beta = 2; \alpha = \beta + 2 = 4.42;$

$(4.42)^2\cos^2\theta = 1 - \sin^2\theta/(2.42)^2;$

$(4.42)^2(1-\sin^2\theta) = (4.42)^2 - (4.42)^2\sin^2\theta$

$= 1 - 0.171\sin^2\theta;\ 19.5 - 1 = (19.5 - 0.17)\sin^2\theta;$

$18.5 = 19.3\sin^2\theta;\ \sin^2\theta = 18.5/19.3 = 0.959;$

$\sin\theta = 0.979;\ \boxed{\theta = 78.3°.}$